Few-Shot SAR Target Classification via Metalearning

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***Abstract*— The state-of-the-art deep neural networks have made a great breakthrough in remote sensing image classification. However, the heavy dependence on large-scale data sets limits the application of the deep learning to synthetic aperture radar (SAR) automatic target recognition (ATR) field where the target sample set is generally small. In this work, a metalearning framework named MSAR, consisting of a metalearner and a base-learner, is proposed to solve the sample restriction problem, which can learn a good initialization as well as a proper update strategy. After training, MSAR can implement fast adaptation with a few training images on new tasks. To the best of our knowledge, this is the first study to solve a few-shot SAR target classification via metalearning. In particular, the few-task problem is defined by analyzing the effect of available training classes on the performance of metalearning models. In order to reduce the metalearning difficulties caused by the few-task problem, three transfer-learning methods are employed, which can leverage the prior knowledge from the pretraining phase. Besides, we design a hard task mining method for effective metalearning. Based on the Moving and Stationary Target Acquisition and Recognition (MSTAR) data set, a specialized data set named NIST-SAR is devised to train and evaluate the proposed method. The experiments on NIST-SAR have shown that the proposed method yields better performances with the largest absolute improvements of 1.7% and 2.3% for 1-shot and 5-shot, respectively, over the next best, which indicates that the proposed method is promising and metalearning is a feasible solution for few-shot SAR ATR.**

***Index Terms*— Automatic target recognition (ATR), few-shot, metalearning, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) plays an important role in the military and civil fields with the all-day and all-weather capability. The SAR target images can provide unique information by capturing the electromagnetic scattering characteristics of the targets, which are important for the SAR target recognition. However, human recognition of SAR targets is challenging and inefficient due to its complicated characteristics. Therefore, the SAR automatic target recognition (ATR) has received intensive attention and many relevant methods are proposed gradually. A standard SAR ATR process can be divided into three steps: detection, discrimination, and classification. The first two steps aim to locate the potential targets and filter the false alarms. In the last step, a classifier is used to recognize the potential targets.

There are many well-designed ATR classifiers for the last classification step, which can be divided into three categories: template-based, model-based, and machine learning-based methods. The semiautomated image intelligence processing system developed by Lincoln Laboratory is a typical template-based example, which maintains massive storage of templates and employs template matching technique to find the best match between the potential targets and the templates [1]–[3]. It is challenging to acquire various templates from SAR targets with different imaging parameters, target poses, and pitch angles, which deteriorates the classification performance of the targets in extended operating conditions. To overcome this inadequacy, the model-based methods [4]–[7] were developed. For example, Moving and Stationary Target Acquisition and Recognition (MSTAR) system only stores the physical models of targets instead of directly storing the target templates. The SAR image features can be online predicted by simulating the scattering process with the help of a computer-aided design model and electromagnetic simulation software. However, the model construction and online prediction will increase the computation burden. Therefore, a tradeoff between the recognition performance and the system complexity is necessary.

With the development of machine learning, the algorithms that can produce large-margin classifiers, such as support vector machine (SVM) and AdaBoost, etc., are employed for SAR ATR in [8] and [9]. Srinivas *et al.* [10] propose a two-stage ATR framework by combining the SAR image features with the learned graphical models. As mentioned above, the traditional machine-learning-based ATR methods [8], [9], [11] usually employ handcrafted features, which limits the recognition performance. While Wagner [12] combines the convolutional features and SVM for SAR ATR, which can implement better performances. Furthermore, Morgan [13] and Chen *et al.* [14] apply convolutional

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neural networks to SAR ATR and achieve the state-of-the-art performance due to the better representative ability of deep learning models.

Although the current SAR target classification models have achieved great success, they are trained by hundreds of thousands of iterations based on large-scale data sets. However, it is a challenge to obtain large-scale target samples with fine annotation for SAR classification due to the following reasons: 1) the data acquisition of SAR is more difficult than that of the natural scene images and 2) the manual interpretation of SAR images is very time-consuming and laborious. The insufficient training examples will limit the performance and development of SAR target classification.

Enlarging training sample sets [15]–[24] is a straightforward method to solve the sample restriction problem from the perspective of data, including the traditional data augmentation operations, the automatic data augmentation and generating new samples by simulators or generative models. Transfer learning [25]–[31] is another feasible approach, it can utilize the prior knowledge learned from the source domain with sufficient training data to solve the sample restriction problem in the target domain. Unsupervised learning can discover information effectively without supervision and it mainly deals with the unlabeled data [29], [32].

In order to reduce the dependence on a large number of training examples, the few-shot learning was proposed and attracted increasing attention in image classification, regression, and reinforcement learning [33]–[37]. However, the above methods of enlarging training data, transfer learning, and unsupervised learning are not the extreme few-shot SAR target classification defined in most of the current few-shot learning researches. This article aims to solve the problem of extreme few-shot SAR target classification via metalearning, where we only have a few (1 or 5) training samples for each SAR target category.

As a canonical few-shot learning method, metalearning [38]–[40] has promoted the relevant researches, which is able to learn the update function or the learning rule in the previous works. In [33], the metalearning model is designed to learn a transformation from the models trained on a few examples to those trained on a large set of examples. The recent metalearning models usually aim to implement few-shot recognition as well as fast adaptation for new targets, e.g., updating networks with few examples and iterations. Ravi and Larochelle [34] learned a long short-term memory based optimizer to train the base-learner network in the few-shot regime, so that the model can converge quickly with a few examples. Differently, Munkhdalai and Yu [35] employed memory-augmented model to learn the metalevel knowledge for fast parameter adaptation on new tasks with a few examples. Besides, model agnostic metalearning (MAML) [36] learns a good initialization, based on which the model can converge quickly for a new few-shot task. Furthermore, metastochastic gradient descent (MSGD) [37] not only learns a good initialization but also a proper learning rate for each parameter of the base-learner.

Most of the current metalearning classification research studies focus on natural scene images. However, to the best of our knowledge, there are limited relevant metalearning ATR methods applied in the remote sensing images, let alone the field of SAR ATR. Considering the characteristic of metalearning, we introduce metalearning to solve the sample restriction problem and realize few-shot SAR target classification.

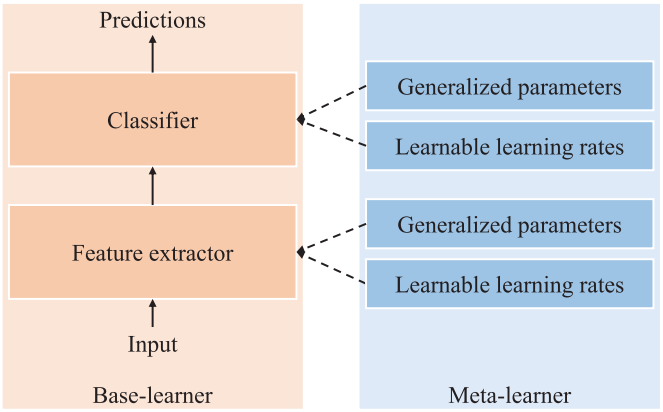


Fig. 1. Metalearning framework consists of a metalearner and a base-learner. The base-learner can be regarded as a standard SAR target classification network. The metalearner stores the generalized parameters and learnable learning rates for feature extractor and classifier of the base-learner.

It is challenging to directly apply the current metalearning methods in natural scene image classification to the SAR target classification. The relevant metalearning methods need to train the models on a metatraining data set with relatively sufficient classes, such as MiniImageNet [34] (64 classes for training) and Omniglot [41] (1200 classes for training), to acquire strong prior knowledge and better generalizing ability. Whereas, the common SAR ATR data sets, e.g., MSTAR, usually have far fewer classes than MiniImageNet and Omniglot, which hardly satisfy the training requirement of metalearning.

To better understand the challenge from fewer classes of SAR and apply the metalearning methods to SAR target classification more effectively, we analyze the effects of training classes on metalearning models and define it as a few-task problem. We illustrate this issue in Section III-B in detail and provide the experimental analysis in Section IV-E. Even though there are almost equal training images, the generalizing abilities of the metalearning models will degrade with fewer training classes. The proposal of the few-task problem can also promote the research studies on other few-shot fields.

In this article, we address the few-shot SAR target classification problem via metalearning. First, a metalearning framework is introduced for a few-shot SAR target classification, which consists of a metalearner and a classifier. The metalearning framework is shown in Fig. 1 and the metalearning process is shown in Fig. 2. This model can learn a good initialization and a proper update strategy from the distribution of many few-shot classification tasks, based on which the trained metalearner can guide the classifier to converge well and quickly. To the best of our knowledge, this is the first study to solve a few-shot SAR target classification

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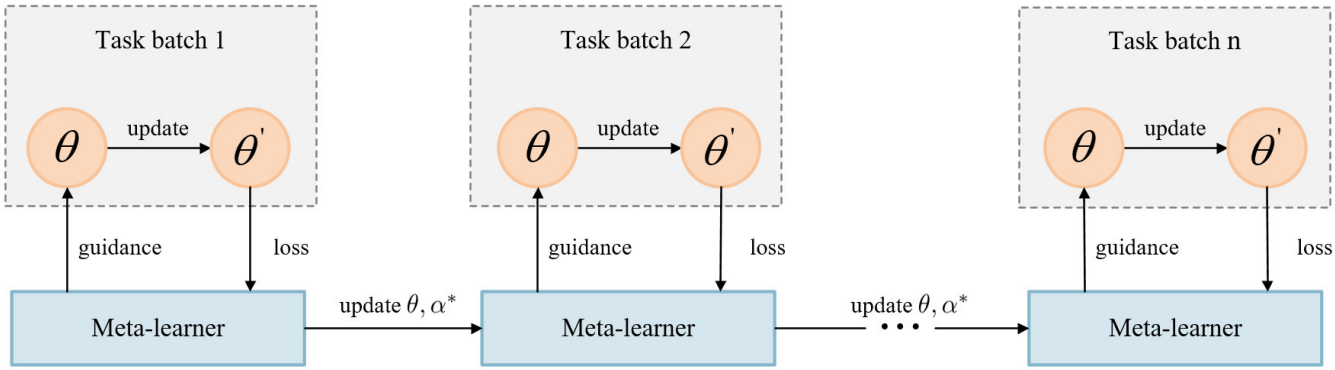


Fig. 2. Metalearning process. θ denotes the generalized initialization parameters of the base-learner. α^* denotes the metalearning rate of each parameter in the base-learner. The metalearner can guide the base-learner to update its parameters by leveraging the training examples on the training subset of each task. Then, the updated parameters θ' is evaluated on the test subset and the loss is used to update the metalearner.

via metalearning. Second, we use three transfer-learning methods to address the few-task problem and alleviate the difficulties of metalearning with fewer training classes. The transfer-learning methods employed in this work can also be used for other small-scale data sets in remote sensing or natural scene images. Third, a hard task mining method is designed to promote the effectiveness of metalearning by making the model pay more attention to the harder tasks. Besides, we partition the MSTAR data set for effective training and evaluation of metalearning. For convenience, we denote the partitioned MSTAR data set as NIST-SAR (Section IV-A2), which can provide convenience for future researches on a few-shot SAR target classification.

We explore the feasibility of metalearning SAR target classification on NIST-SAR. As mentioned above, this is harder than that on MiniImageNet or Omniglot data set due to fewer target categories. The experiment results on the NIST-SAR data set have proved the effectiveness of the proposed method. With the help of the transfer-learning methods, the meta-learning model yields better performance in the 1-shot and 5-shot settings, and the hard task mining method can significantly improve the few-shot classification performance.

The remainder of this article is organized as follows. Section II gives a preliminary introduction about the problem setup, data organization, the definition of task and process of metalearning. The detailed metalearning pipeline, the few-task problem, the transfer-learning methods, the hard task mining method and the framework implementation are successively described in Section III. The NIST-SAR data set and experiment results are presented in Section IV. Section V gives the conclusions.

II. Preliminary

In this section, we introduce the problem setup, data organization, the definition of task and process of metalearning, following the relevant works [36], [37].

A. Problem Setup

Metalearning, also known as *learning to learn*, aims to learn how to learn on a new task. After training on a series of few-shot tasks $\{\mathcal{T}_i\}$ (metatraining tasks) from the distribution $p(\mathcal{T})$, the metalearning models can implement fast adaptation with few training examples on new tasks (metatest tasks).

Specifically, the whole metalearning framework usually consists of a metalearner and a base-learner. The metalearner is responsible to guide the update of the base-learner for quick and good convergence on a few-shot task. In general, the base-learner is a standard deep learning model, e.g., a convolutional neural network for image classification, which is updated quickly on each task with the guidance of the metalearner. Both metalearner and base-learner are integrated into an end-to-end framework for effective training. Once trained on metatraining tasks, the metalearner can teach the base-learner how to update quickly on metatest tasks. We introduce the relevant concepts in metalearning in the following.

B. Data Organization

The metalearning data set is composed of metatraining set $\mathcal{D}^{\text{tr}}$ and metatest set $\mathcal{D}^{\text{te}}$, as shown in Fig. 3. $\mathcal{D}^{\text{tr}}$ only contains metatraining classes, and $\mathcal{D}^{\text{te}}$ only contains metatest classes. These two sets are disjoint, i.e., $\mathcal{D}^{\text{tr}} \cap \mathcal{D}^{\text{te}} = \emptyset$. This setting can ensure the metalearning model is evaluated on the new tasks of metatest classes. As shown in Fig. 3, the data of metatraining tasks are sampled from $\mathcal{D}^{\text{tr}}$ and the data of metatest tasks are sampled from $\mathcal{D}^{\text{te}}$.

C. Definition of Task

As shown in Fig. 3, a task contains two subsets: training subset $\mathcal{T}^{\text{tr}}$ and test subset $\mathcal{T}^{\text{te}}$. An N-way, K-shot task can be defined by the number of classes N, and the number of samples per class K in the training subset $\mathcal{T}^{\text{tr}}$, which can be formulated as follows:

$$\text{Task}_{N-K}: \begin{cases} \text{Sample } N \text{ categories}, & N \leq C \\ \text{Sample } K \text{ training examples per class} \end{cases} \tag{1}$$

C is the number of target categories on the whole metatraining or metatest data set. For an N-way, K-shot task, we first sample N classes, then randomly select K samples for each of N classes to construct training subset $\mathcal{T}^{\text{tr}}$, i.e., there are total $N \times K$ samples in training subset $\mathcal{T}^{\text{tr}}$, and randomly

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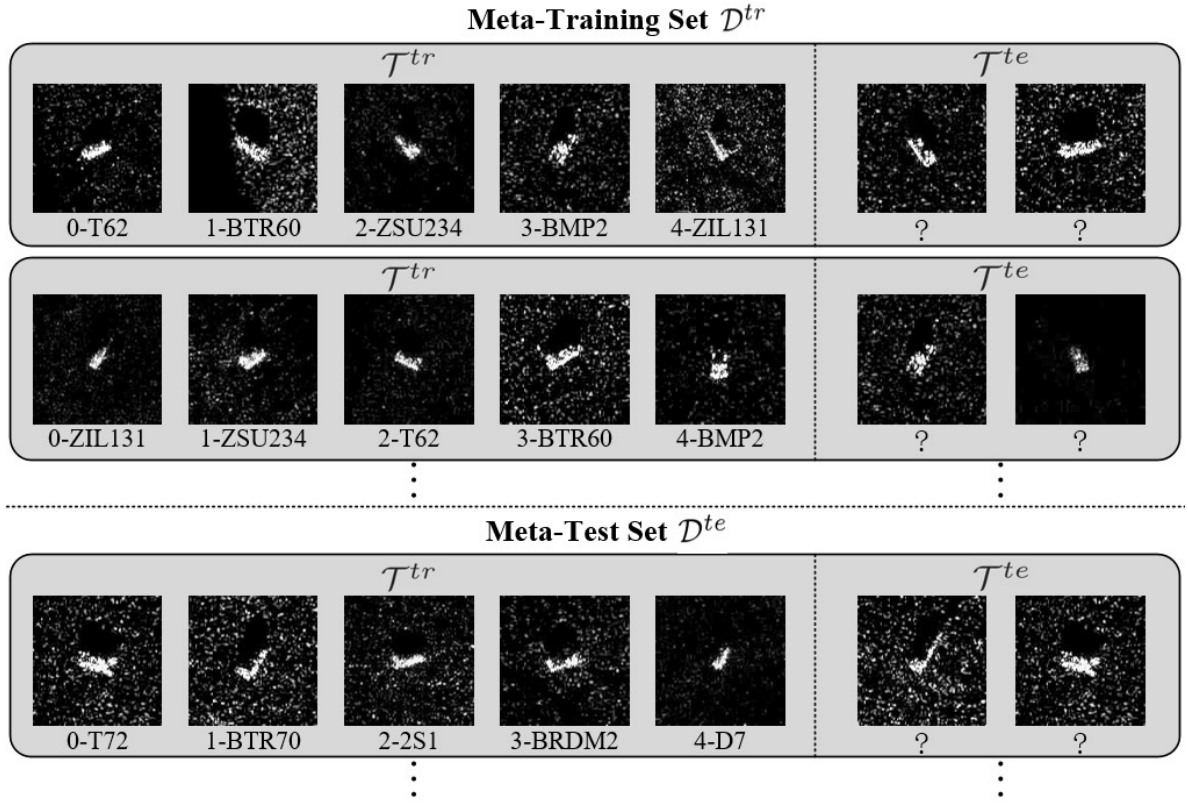


Fig. 3. 5-way 1-shot SAR ATR setup. There are metatraining set $\mathcal{D}^{\text{tr}}$ and metatest set $\mathcal{D}^{\text{te}}$ on the proposed NIST-SAR data set. Within each task (the gray box), there are two subsets: training subset $\mathcal{T}^{\text{tr}}$ and test subset $\mathcal{T}^{\text{te}}$, they have same target classes and different images. For 5-way, 1-shot, we sample one image per class for $\mathcal{T}^{\text{tr}}$, $\mathcal{T}^{\text{te}}$ on $\mathcal{D}^{\text{tr}}$ and $\mathcal{T}^{\text{tr}}$ on $\mathcal{D}^{\text{te}}$, while 15 images per class are randomly selected for $\mathcal{T}^{\text{te}}$ on $\mathcal{D}^{\text{te}}$. The classes on $\mathcal{D}^{\text{tr}}$ are not present on $\mathcal{D}^{\text{te}}$.

select a number of samples of the same N classes to construct test subset $\mathcal{T}^{\text{te}}$. $\mathcal{T}^{\text{tr}}$ and $\mathcal{T}^{\text{te}}$ have the same target categories but different samples. In few-shot learning, K is a very small value (K is 1 or 5 in this article). In this article, we define the tasks with different arrangements of SAR target categories as different types.

D. Metatraining Phase

In each task, we first train the model on $\mathcal{T}^{\text{tr}}$, and then update the parameters of the base-learner from θ to θ' under the guidance of metalearner, so that we have a temporary base-learner for the current task. The number of update on $\mathcal{T}^{\text{tr}}$ is usually very small, e.g., 1 in [37], to implement fast parameter adaptation and avoid overfitting. We follow the setting of 1-iteration update on $\mathcal{T}^{\text{tr}}$ in this work. The updated base-learner is then tested on $\mathcal{T}^{\text{te}}$. Here, θ is the goal of metalearning, i.e., the generalized initialization parameters, whereas the updated parameters θ' are not preserved, which are temporary on the current task. The losses from $\mathcal{T}^{\text{te}}$ of a batch of tasks are used to update θ.

E. Metatest Phase

During metatest phase, the metalearning model is evaluated on metatest tasks $\mathcal{T}_{\text{metatest}}$ of metatest classes. For each task, the metalearning model is first trained on $\mathcal{T}^{\text{tr}}$ with a few examples per class, and then evaluated on $\mathcal{T}^{\text{te}}$. The test sets on metatest tasks have more test examples (15 images per class in this work) than metatraining tasks for effective evaluation. The average result on $\{\mathcal{T}_i^{\text{te}}\}$ of all metatest tasks is the final evaluation.

In summary, there are some differences between metalearning and supervised learning.

1) The data sets for metalearning, e.g., MiniImageNet divided into $\mathcal{D}^{\text{tr}}$ and $\mathcal{D}^{\text{te}}$, are different from those in supervised learning, e.g., ImageNet [42]. Hence, the target categories for metatraining and metatesting are different.
2) The fundamental unit for training and testing the metalearning models is *task*. The metalearning model is trained on a series of metatraining tasks $\mathcal{T}_{\text{metatrain}}$. Then, it is evaluated on the metatest tasks $\mathcal{T}_{\text{metatest}}$.

It should be noted that although metalearning is used to address the few-shot problem, the metalearning models still need to be trained on sufficient data, i.e., $\mathcal{D}^{\text{tr}}$.

III. Metalearning for Few-Shot SAR Target Classification

In this section, we first introduce the metalearning pipeline, and then define the few-task problem. To tackle the few-task problem, three transfer-learning methods are introduced. Furthermore, the hard task mining method is proposed to promote the effective learning of the metalearning model. Finally, we describe the detailed implementation of the metalearning framework.

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A. Metalearning Pipeline

In supervised deep learning, the models are trained on large-scale data sets for hundreds of thousands of iterations by SGD. Although there are a few training examples on each minibatch, the large-scale training data can ensure there are sufficient mini-batches to train the model. However, in extremely low-data regime, there are only a few even one minibatch to train the model. Therefore, the optimization strategies used in standard supervised deep learning may not work well for few-shot learning. Besides, in a few-shot regime, the deep learning-based models can only learn limited knowledge from scarce examples, which makes a good initialization very crucial. Based on this analysis, a good initialization and proper update strategy are important in a few-shot regime. Inspired by the relevant metalearning methods in natural scene image recognition, the proposed model learns a good initialization and a proper update strategy for a few-shot SAR ATR, shown in Algorithm 1.

For supervised learning ATR, the model f is trained based on the empirical loss $\mathcal{L}_{\mathcal{T}}$ by SGD as the following equation:

$$\theta' = \theta - \alpha \nabla \mathcal{L}_{\mathcal{T}}(\theta) \tag{2}$$

where α is the learning rate for SGD. The model parameters are updated to θ' by leveraging the feedback from the parameters θ. The empirical loss $\mathcal{L}_{\mathcal{T}}$ is usually computed as

$$\mathcal{L}_{\mathcal{T}}(\theta) = \frac{1}{|\mathcal{T}^{\text{tr}}|} \sum_{(x,y)\in\mathcal{T}^{\text{tr}}} (\ell(f_\theta(x), y)). \tag{3}$$

In this work, we employ the cross-entropy loss as the loss function ℓ for SAR ATR. x and y are the input image and class label, respectively. $|\mathcal{T}^{\text{tr}}|$ is the number of training examples on $\mathcal{T}$.

For few-shot SAR ATR, the proposed metalearning model is designed to learn the learning rate α^*, so that it can update the parameters θ automatically. This setting can make the metalearner approach the optimal update strategy to avoid overfitting or underfitting. Therefore, the SGD process in (2) can be reformulated as

$$\theta' = \theta - \alpha^* \nabla \mathcal{L}_{\mathcal{T}}(\theta) \tag{4}$$

where α^* is the learnable learning rate and it is not manually defined. The metalearner can learn α^* from the task distribution $p(\mathcal{T})$ in the metatraining phase. α^* is set for each parameter of the base-learner as in [37].

Mathematically, the objective of the proposed metalearning model can be formulated as follows:

$$\min_{\theta,\alpha^*} \sum_{\mathcal{T}_i\sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'}) = \min_{\theta,\alpha^*} \sum_{\mathcal{T}_i\sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta-\alpha^*\nabla\mathcal{L}_{\mathcal{T}_i}(f_\theta)}) \tag{5}$$

where the parameters θ' are for the temporary base-learner on the current task.

The proposed metalearning model aims to learn a good metalearner with a good initialization θ and the proper update strategy α^*, which can guide the base-learner to get the temporary parameters θ' for the current task. Hence, the metalearner should be trained based on the updated parameters θ' in (5).

With the above analysis, we can train the metalearning model by SGD just like the one in the supervised learning. The loss $\mathcal{L}_{\mathcal{T}}$ for SGD is from the updated parameters θ'. This process can be formulated as follows:

$$(\theta, \alpha^*) \leftarrow (\theta, \alpha^*) - \beta \nabla \sum_{\mathcal{T}_i\sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'}) \tag{6}$$

where β is the manually defined learning rate to train the metalearner.

Algorithm 1 Metalearning Pipeline

Input: Few-shot SAR ATR task distribution $p(\mathcal{T})$, learning rate β, the classification model f

Output: good initialization parameters θ, learnable learning rate α^*

Initialize θ, α^*, I iterations
for $i \leftarrow 1; i \leq I$ **do**
  Sample n tasks from $p(\mathcal{T})$
  for $j \leftarrow 1; j \leq n$ **do**
    $\mathcal{L}_{\mathcal{T}_j^{\text{tr}}} \leftarrow \frac{1}{|\mathcal{T}_j^{\text{tr}}|} \sum_{(x,y)\in\mathcal{T}_j^{\text{tr}}} \ell(f_\theta(x), y)$
    $\theta' \leftarrow \theta - \alpha^* \nabla \mathcal{L}_{\mathcal{T}_j^{\text{tr}}}$
    $\mathcal{L}_{\mathcal{T}_j^{\text{te}}} \leftarrow \frac{1}{|\mathcal{T}_j^{\text{te}}|} \sum_{(x,y)\in\mathcal{T}_j^{\text{te}}} \ell(f_{\theta'}(x), y)$
    $j \leftarrow j + 1$
  end for
  $(\theta, \alpha^*) \leftarrow (\theta, \alpha^*) - \beta \nabla \sum_{j\in[1,n]} \mathcal{L}_{\mathcal{T}_j^{\text{te}}}$
  $i \leftarrow i + 1$
end for

B. Few-Task

As illustrated in Section II, different from supervised learning, the fundamental training unit for metalearning is *task*. For supervised learning, the models can acquire generalized knowledge from a large number of training examples. Similarly, the metalearning models need to be trained on as many tasks as possible for better performance and generalizing ability. Hence, the analysis of the metalearning task is essential.

According to the definition of task in Section II, the number of N-way, K-shot tasks is equal to the number of permutations. We define the tasks with different arrangements of SAR target categories as different types. For a data set with C target categories, there are A_C^N different tasks. It can be formulated as follows:

$$A_C^N = C \times (C-1) \times (C-2) \cdots (C-N+1). \tag{7}$$

The common SAR ATR data sets, e.g., MSTAR, have fewer target categories than those in the natural scene image classification, which brings about the problem of few-task for metalearning SAR ATR. We analyze the number of some metalearning data sets in the natural scene and SAR ATR, shown in Fig. 9. Apparently, there are fewer tasks for metalearning SAR ATR. Based on this observation, we define the few-task problem.

Definition 1: Few-Task: In the metalearning regime, the few-task problem is that there are hundreds or even fewer

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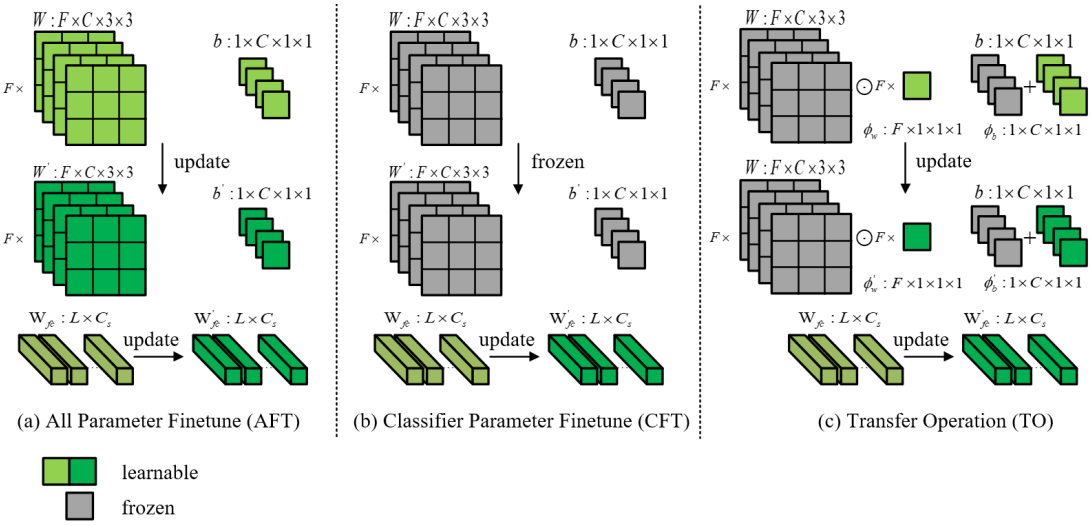


Fig. 4. Transfer-learning pipeline, the network is updated based on the pretrained parameters. (a) AFT updates all parameters of the model, it may cause overfitting with few training examples. (b) CFT is just updating the parameters of the classifier (the last fully connected layer) to alleviate overfitting. (c) TO can reduce the number of learning parameters by freezing the parameters of the convolutional layers and only learning the additional parameters ϕ_w and ϕ_b. It can also preserve the prior knowledge learned from the large-scale data set. It only sets one learning parameter ϕ_w for each convolutional filter.

training tasks due to the limitation of the number of target categories.

The existing metalearning data sets, such as Omniglot and MiniImageNet, have dozens of target categories to avoid the few-task problem. We argue that the few-task problem is nontrivial for metalearning researches both in natural scene images and SAR ATR for the following reasons.

1) The few-task problem may be inevitable in some metalearning applications, e.g., SAR ATR, where there are no sufficient target categories.
2) The few-task problem can lead to worse performance and overfitting for metalearner, as shown in the experiment in Section IV-E.

An Intuitive Understanding: Here, the *task* in metalearning can be compared to the training examples in supervised learning. The supervised learning models will suffer from overfitting with insufficient training examples. Similarly, the overfitting problem can also occur for metalearning model with insufficient training *tasks*. Some data augmentation operations of increasing the number of training images such as random cropping, mirroring, and resizing, can alleviate even avoid overfitting. But for the few-task problem, the task augmentation may not work because increasing the number of training classes is extremely difficult, which is of great significance to study this problem.

C. Transfer-Learning Methods

As the above analysis, this work needs to address the few-task problem for SAR ATR except few-shot learning. In supervised learning, the insufficient training examples will lead to overfitting. This is because compared with the limited knowledge from few examples, the deep models with a large number of learning parameters are too complicated [43]. Hence, the few-shot learning models [36], [37], [43], [44] usually have fewer parameters to avoid overfitting.

Similarly, fewer tasks can also cause overfitting for metalearning. We use transfer-learning to alleviate the difficulties of metalearning. As shown in Fig. 4(a), a straightforward method of transfer-learning named AFT is pretraining the base-learner on the metatraining set $\mathcal{D}^{\text{tr}}$ in the supervised learning manner, which is easy to train, to acquire a better initialization, then, finetuning the parameters of the pretrained base-learner with metalearning.

Although the model of AFT can acquire some prior knowledge by pretraining, the number of learning parameters is still too large. Furthermore, a reasonable solution for few-task problem is reducing the number of learning parameters. As shown in Fig. 4(b), a simple transfer-learning method called CFT is to freeze the parameters of the feature extractor and only finetune the last fully connected layer after pretraining the base-learner on the metatraining set $\mathcal{D}^{\text{tr}}$. This operation can reduce the learning parameters compared with AFT.

However, only training the last fully connected layer will increase the difficulties of the adaptation to the data distribution on the target data set. In addition, too few learning parameters can also degrade the performance of the metalearning models. Inspired by the study [45], we introduce the transfer operation (TO), shown in Fig. 4(c). This operation incorporates the thoughts of transfer-learning [46] and dictionary learning [43]. Details of TO are described in the following.

We first train the base-learner on metatraining set $\mathcal{D}^{\text{tr}}$ to acquire a good prior knowledge, as follows:

$$(\Theta', \mu') = (\Theta, \mu) - \alpha \nabla \mathcal{L}_{\mathcal{D}^{\text{tr}}}((\Theta, \mu)). \tag{8}$$

Here, the base-learner is composed of a feature extractor Θ (the convolutional layers) and a classifier μ (the last fully connected layer).

Then the parameters $\phi = (\phi_w, \phi_b)$ for TO are learned in the metalearning manner on SAR ATR data set, as shown in Fig. 4(c). In [45], the pretrained parameters of the feature extractor Θ are frozen in the metalearning phase,

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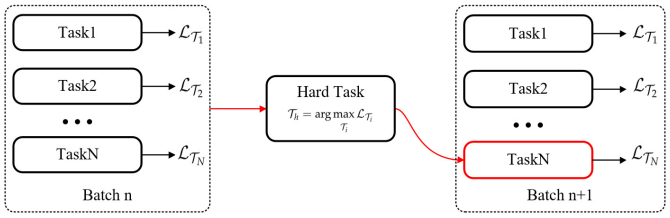


Fig. 5. Pipeline of the proposed hard task mining method.

the parameters ϕ_w are set for each channel of all convolutional filters in the base-learner. The proposed TO only sets one learnable parameter for each convolutional filter in the base-learner. This operation can further reduce the number of learning parameters. We set a learnable learning rate α^* for each parameter of TO to learn a better update strategy.

Hence, the learning parameters of the base-learner are ϕ and μ. This process can be formulated as

$$(\mu, \phi, \alpha^*) \leftarrow (\mu, \phi, \alpha^*) - \beta \nabla \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f([\Theta, \mu', \phi'])). \quad (9)$$

In (9), ϕ consists of ϕ_w and ϕ_b, which are set for the weights and biases of the convolutional layers, respectively. The parameters μ' of the classifier are updated by

$$\mu' = \mu - \alpha^* \nabla \mathcal{L}_{\mathcal{T}}([\Theta, \mu, \phi]). \quad (10)$$

And ϕ' is updated by

$$\phi' = \phi - \alpha^* \nabla \mathcal{L}_{\mathcal{T}}([\Theta, \mu, \phi]). \quad (11)$$

We denote a convolutional layer with weight W and bias b as (W, b), and X is input. For a standard convolutional operation

$$\{(W, b), X\} = W * X + b. \quad (12)$$

If we apply ϕ to (W, b), the above equation will be reformulated as

$$\{(\phi_w, \phi_b), (W, b), X\} = (W \odot \phi_w)X + (b + \phi_b). \quad (13)$$

Here, $\odot$ denotes the elementwise multiplication.

In summary, we introduce three different transfer-learning methods based on the pretrained parameters on the metatraining set $\mathcal{D}^{\text{tr}}$ to address the few-task problem of metalearning, which are as follows.

1) *AFT:* This method finetunes all pretrained parameters of the base-learner during metalearning.
2) *CFT:* This method freezes the parameters of the feature extractor and finetunes the parameters of the classifier (last fully connected layer) during metalearning.
3) *TO:* This method freezes the parameters of feature extractor and adds additional learning parameters for each convolutional filter, then finetunes the additional parameters and the classifier during metalearning.

D. Hard Task Mining

We propose a simple and effective hard task mining method, shown in Fig. 5, to promote the training of the metalearning model. The few training examples and tasks can only provide limited knowledge. The efficient way of using the limited training data is crucial for few-task metalearning. We aim to make the metalearning model pay more attention to the harder tasks during training.

Within each task, the proposed model is first trained on $\mathcal{T}^{\text{tr}}$. The loss from $\mathcal{T}^{\text{tr}}$ is used to update the parameters of the base-learner. Hence, we can get a temporary base-learner on the current task, and test it on $\mathcal{T}^{\text{te}}$. We estimate the difficulty of the task according to the loss $\mathcal{L}_{\mathcal{T}_i}$ on $\mathcal{T}^{\text{te}}$. For a batch of tasks, we simply select one hard task $\mathcal{T}_h$ with the highest loss value on $\mathcal{T}^{\text{te}}$, as the following equation:

$$\mathcal{T}_h = \arg\max_{\mathcal{T}_i} \mathcal{L}_{\mathcal{T}_i}. \quad (14)$$

If the hard task is added, we will select $N-1$ tasks to ensure the number of tasks is N for every batch.

During the metatraining phase, the hard task mining is used to force the metalearning model to learn knowledge effectively. The hard task from the last batch is added to the current batch. We do not employ hard task mining during the metatest phase.

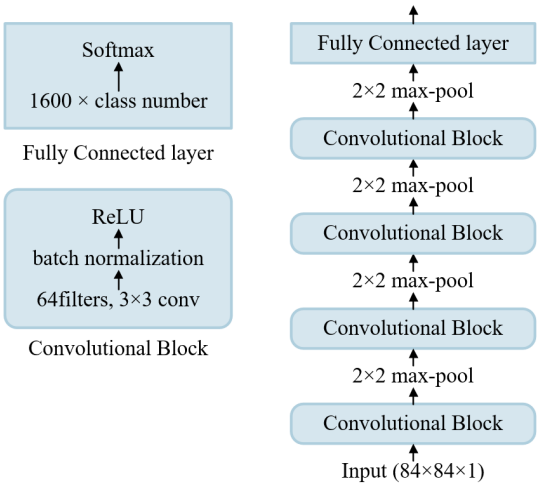


Fig. 6. Architecture of the 4CONV neural network. As illustrated in Section III-C, we apply three different transfer-learning methods to this network.

E. Implementation

1) Model Architecture: We implement the proposed metalearning framework based on the 4CONV neural network, which is commonly used in [36] and [37]. 4CONV, shown in Fig. 6, consists of four convolutional blocks for feature extraction and a fully connected layer with Softmax as the classifier. In each convolutional block, there are a convolutional layer with 3×3 convolution and 64 filters, batch normalization (BN), ReLU nonlinearity and 2×2 max-pooling. We set the learnable learning rate α^* for each learning parameter of the model.

2) Pipeline: The whole pipeline can be divided into three steps: pretraining, metatraining and metatest.

1) *Pretraining:* The 4CONV neural network is trained on the metatraining data set $\mathcal{D}^{\text{tr}}$ in the supervised learning manner. This step is to acquire prior knowledge for transfer-learning methods in the following metatraining phase.

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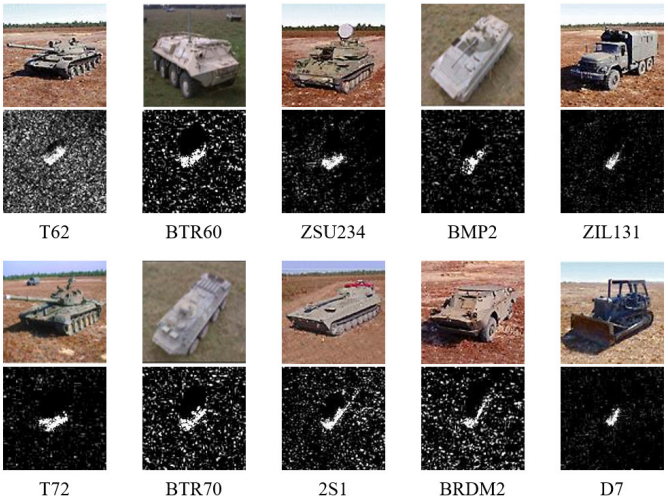


Fig. 7. (Top) Optical images and (Bottom) SAR images for military targets in MSTAR data set. The SAR images are in 17° depression angle.

2) *Metatraining:* In this step, we train the model with transfer-learning methods and hard task mining in a metalearning manner. For AFT, all parameters of four convolutional blocks and the classifier are finetuned. For CFT, only the parameters of the classifier are finetuned. For TO, only the parameters of TO and the classifier can be finetuned. Besides, the learnable learning rates α^* are learned for the three methods during metatraining phase.
3) *Metatest:* This step is to evaluate the trained metalearning model, which has learned a good initialization and a proper update strategy. The base-learner is updated by leveraging the loss of $\mathcal{T}^{\text{tr}}$, then evaluated on $\mathcal{T}^{\text{te}}$.

IV. Data Set and Experiments

A. Data Set

1) MSTAR: The collection of MSTAR data set is jointly supported by the Defense Advanced Research Projects Agency and the Air Force Research Laboratory. Fig. 7 shows some SAR images and the corresponding optical images in this data set. There are ten released military targets (T62, BTR60, ZSU234, BMP2, ZIL131, T72, BTR70, 2S1, BRDM2 and D7), which are collected by an X-band SAR sensor.

2) NIST-SAR: Based on the MSTAR data set, we devised the NIST-SAR data set for metalearning SAR ATR by following the MiniImageNet [47] data set in few-shot image classification. The NIST-SAR data set consists of various few-shot tasks based on the MSTAR data set according to different settings, which can be used to train and evaluate the metalearning and other related few-shot SAR target classification methods. The NIST-SAR data set contains ten military targets in 17° depression angle. It can be divided into $\mathcal{D}^{\text{tr}}$ (metatrain) and $\mathcal{D}^{\text{te}}$ (metatest), as shown in Fig. 3. There are five target categories in $\mathcal{D}^{\text{tr}}$ as metatraining classes and five target categories in $\mathcal{D}^{\text{te}}$ as metatest classes. The details of NIST-SAR are shown in Table I. This setting is to mimic the fact that there are insufficient examples for some targets as well as make the metalearning problem at an appropriate difficulty level.

TABLE I
Details of the Proposed NIST-SAR Data Set

	Class	Serial No.	Depression	No.Image
$\mathcal{D}^{tr}$	T62	a51	17	299
	BTR60	k10yt7532	17	256
	ZSU234	d08	17	299
	BMP2	9566	17	232
	ZIL131	e12	17	299
$\mathcal{D}^{te}$	T72	132	17	232
	BTR70	c71	17	233
	2S1	b01	17	299
	BRDM2	e71	17	298
	D7	92v13015	17	299

B. Baseline

The relevant works for few-shot SAR ATR are so few that we compare the proposed method with traditional machine learning algorithms, e.g., k-nearest neighbor (kNN, k is the number of the used nearest neighbors), and the state-of-the-art metalearning algorithms in natural scene image classification such as MAML [36] and MSGD [37]. Besides, we compare the proposed model with a transfer-learning approach.

C. Experimental Settings

The metalearning framework with TO is denoted as MSAR_{TO} in the following. We apply AFT and CFT to the metalearning models MAML and MSGD and apply TO to MSGD due to its better performance, these models are denoted as the following.

1) $MAML_{AFT}$: Pretrain MAML on the metatraining set of NIST-SAR, then finetune all its parameters on the metatraining set of NIST-SAR.
2) $MAML_{CFT}$: Pretrain MAML on the metatraining set of NIST-SAR, then finetune the parameters of its classifier on the metatraining set of NIST-SAR.
3) $MSGD_{AFT}$: Pretrain MSGD on the metatraining set of NIST-SAR, then finetune all its parameters on the metatraining set of NIST-SAR.
4) $MSGD_{CFT}$: Pretrain MSGD on the metatraining set of NIST-SAR, then finetune the parameters of its classifier on the metatraining set of NIST-SAR.
5) $MSAR_{TO}$: Pretrain MSGD on the metatraining set of NIST-SAR, then finetune the parameters of TO and the classifier on the metatraining set of NIST-SAR.

The transfer-learning baselines are as follows.

1) TF_{AFT100}: We fine-tune all parameters of the base-learner for 100 epochs within each few-shot task.
2) TF_{CTF100}: We fine-tune the parameters of the classifier of the base-learner for 100 epochs within each few-shot task.
3) TF_{AFT1}: We fine-tune all parameters of the base-learner for one epoch within each few-shot task.

The pretraining phase is in supervised learning manner, we use the Adam optimizer with learning rate of 0.001, 16 batch sizes, and 10 000 iterations. Then, the metatraining

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TABLE II

SETTINGS OF DIFFERENT N-WAY K-SHOT SAR ATR. THERE ARE $N \times K$ IMAGES FOR EACH SUBSET EXCEPT $N \times 15$ IMAGES FOR THE TEST SET OF METATEST

setting	meta-train		meta-test	
	train	test	train	test
5-way 1-shot	5×1=5	5×1=5	5×1=5	5×15=75
5-way 5-shot	5×5=25	5×5=25	5×5=25	5×15=75

is executed. For AFT, all parameters of the base-learner and the learnable learning rates are learned during metatraining. For CFT, the pretrained parameters of the feature extractor are fixed and the model only learns the parameters of the classifier and the learnable learning rates. For TO, the pretrained parameters of the feature extractor are fixed and the model only learns the parameters of the classifier and TO, and the learnable learning rates. The cosine annealing scheduling is applied on the Adam optimizer with a learning rate of 0.001. We metatrain the models for 10 000 episodes and each episode contains four inner tasks (that is, the batch of tasks is 4).

We metatest the model every 200 episodes and record the highest test accuracies. In each metatest phase, there are 200 test tasks to evaluate the trained model with 1000 (200×5) training images and 15 000 (200×75) test images for 5-way, 1-shot and 5000 (200×25) training images and 15 000 (200×75) test images for 5-way, 5-shot. We metatrain and metatest the metalearning baselines in the same way. There are 200 metatest tasks to evaluate the kNN algorithms.

The number of training and test images for metatraining and metatest are listed in Table II. For example, in a 5-way 1-shot task, five categories are sampled in any order, then one example is selected for each class. Hence, there are five images for the training set. For the test set of metatraining, five images per class are sampled from the remaining. For the metatest set, we sample images in the same way as on the metatraining set except 15 images per class on the test subset for effective evaluation. It should be noted that the image labels in each task are randomly assigned, which means there are $A_5^5 = 120$ different tasks. Therefore, the number of test images and the diversity of tasks are enough for evaluation.

D. Performance

The evaluation results with 95% confidence intervals of the proposed model for 5-way, 1-shot and 5-way, 5-shot are shown in Table III. We also visualize the metatest accuracies every 200 metatraining episodes in Fig. 8. Without hard task mining, the trained model MSAR_{TO} can implement 52.2% accuracy with one training image per unseen class. When increasing the number of training images per class to 5, the proposed model can yield a better performance of 59.0% accuracy. The experiment results indicate the effectiveness of the proposed method and metalearning is a feasible solution for a few-shot SAR ATR.

We compare MSAR_{TO} with kNN, transfer-learning approach, MAML [36] and MSGD [37] in the 5-way, 1-shot and 5-way, 5-shot settings. In addition, we evaluate the performances of meta-learning models MAML_{AFT}, MAML_{CFT}, MSGD_{AFT} and MSGD_{CFT} which are the baselines with better performances.

TABLE III

MEAN ACCURACIES OF THE 5-WAY 1-SHOT AND 5-SHOT TASKS ON THE NIST-SAR DATA SET, WITH 95% CONFIDENCE INTERVALS. AFT: ALL PARAMETER FINETUNE, CFT: CLASSIFIER PARAMETER FINETUNE. FOR EXAMPLE, MAML_{AFT} IS TO PRETRAIN MAML ON THE METATRAINING SET OF NIST-SAR, THEN FINETUNE ALL ITS PARAMETERS ON THE METATRAINING SET OF NIST-SAR. WE DENOTE THE METALEARNING METHOD MSGD WITH TO AS MSAR_{TO}. HT DENOTES THE MODEL WITH HARD TASK MINING METHOD

Method	1-shot	5-shot
kNN(k=5)	20.0±0.0	33.3±0.83
kNN(k=3)	29.3±0.46	36.0±0.92
TF_{AFT100}	35.3±1.10	58.0±0.45
TF_{CFT100}	36.1±0.71	58.2±0.82
TF_{AFT1}	22.5±1.05	19.6±1.21
MAML [36]	30.6±1.32	33.3±1.25
MAML_{AFT}	28.5±1.12	24.4±1.03
MAML_{CFT}	21.2±0.97	22.5±0.91
MSGD [37]	51.8±1.17	49.3±0.73
MSGD_{AFT}	51.3±1.03	56.0±1.23
MSGD_{CFT}	48.3±0.92	53.3±1.06
MSAR_{TO}	52.2±0.98	59.0±1.04
MSAR_{TO+HT}	**53.5±1.0**	**60.5±0.9**

As shown in Table III, $\text{MSAR}_{\text{TO+HT}}$ yields better performances both for 1-shot (53.5%) and 5-shot(60.5%). The metatest accuracy curves are shown in Fig. 8. The state-of-the-art metalearning methods, such as MAML and MSGD, suffer from overfitting and have poor performance. Note that although MSGD can yield 51.8% accuracy for 1-shot, the accuracy declines gradually as the training continues. MAML and MSGD work well for natural scene images, but the few-task problem, commonly existed in SAR ATR, degrades their performances. Another problem is that MSGD suffers from unstable performance, that is, the performance of MSGD for 5-shot is worse than 1-shot, we believe this is because a relatively small number of metatraining classes and images can only provide limited information for the metalearning models with a relatively large number of parameters.

As shown in Table III, the proposed model can achieve better performances than the transfer-learning methods for 1-shot and 5-shot. It should be noted that we only update the parameters of the proposed model with one iteration within each task. The performances of the transfer-learning methods are worse when we finetune the model with only one iteration as ours. This indicates the effectiveness of the proposed metalearning method.

By combining with transfer-learning methods AFT and CFT, MSGD_{AFT} can yield better performances for 5-way 5-shot. Because the pretraining in a supervised-learning manner is easier for MSGD to acquire good knowledge than the metatraining based on different few-shot tasks. For 1-shot, MSGD_{AFT} also suffers from certain overfitting and

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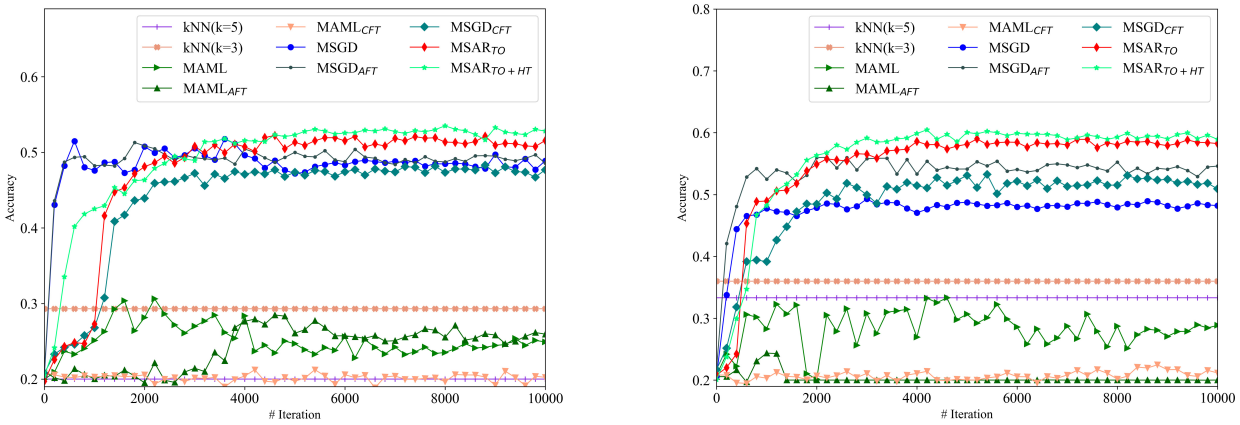


Fig. 8. Experiment results for different methods on NIST-SAR. (a) 5-way, 1-shot. (b) 5-way, 5-shot.

has a similar performance to MSGD. MSGD$_{CFT}$ yields a worse accuracy of 48.3%, because for metalearning, only metatraining the classifier is challenging to transfer the knowledge from metatraining classes to metatest classes.

This experiment can simulate the few-shot situation where some target categories have relatively sufficient training images while others have few training images even one. For SAR ATR, what is more serious is that there are commonly few available target categories with sufficient training images to train the metalearning models. In contrast, for natural scene images, the number of metatraining classes is larger, e.g., 64 metatraining classes on MiniImageNet data set. Therefore, the metalearning models can be trained more effectively and implement better performances for few-shot classification of natural scene images. We define the above phenomenon of few metatraining classes in SAR ATR as the few-task problem in Section III-B. The experiment results show that the proposed method can solve the few-task problem effectively.

The few-task problem, defined by this work, has also important influence on the few-shot researches of other areas. In the real world, we may not always acquire sufficient metatraining classes to train metalearning models, and the few-task problem will cause poor performance. The solution for the few-task problem proposed in this work can also be used for other areas and provide some inspiration for future research studies.

E. Few-Task Analysis

In this section, we analyze the few-task problem in detail. This problem has been described and defined in Section III-B, it is very common in SAR ATR due to the difficulties of the data collection and annotation. As shown in the above section, the few-task problem severely influences the performances of the successful few-shot methods in natural scene images. The number of metatraining classes and tasks is shown in Fig. 9(b) and (c), indicating the insufficiency of metatraining data for SAR ATR.

We design comparison experiments with the 5-way, 1-shot setting to verify the effect of the few-task problem. For a fair comparison, the number of metatraining images for different experiments is set as equal as possible by changing the number of metatraining images per class. The experimental settings are listed in Table IV. The number of target categories is set from 5 to 64. There are about 3200 metatraining images for each experiment. In order to analyze the influence of the number of the metatraining tasks, the experiments are performed on MiniImageNet of relatively more metatraining images and classes. With the above elaborate design, the comparison experiments can expose the few-task problem effectively.

The experimental results are shown in Fig. 10, the model is evaluated on metatest classes of MiniImageNet data set. That is, we train the model with sufficient few-shot tasks on metatraining classes, and evaluate the model with 200 test tasks on metatest classes (unseen classes) every 200 training iterations. With the training on metatraining classes, the model may become overtrained on metatraining classes, resulting in the loss increasing on metatest classes. This is exactly what we want to reveal by this comparison experiment.

Metalearning can avoid the above problem of overtrained and improve the recognition accuracy for novel classes by training on various few-shot tasks. The models are trained with more classes have better performances due to acquiring stronger prior knowledge from the more metatraining classes and tasks. When decreasing the number of metatraining classes, the generalizing abilities of the metalearning models will degrade even though there are almost equal metatraining images.

The few-task problem usually can cause worse performance and it is inevitable in some metalearning applications with insufficient metatraining classes, especially for SAR ATR due to the difficulties of data collection and annotation. Hence, solving this problem well will further expand the application areas of the current metalearning methods.

F. Ablation Study

This work employs TO to reduce the number of the learning parameters and transfer the knowledge from supervised-learning to metalearning on metatraining classes.

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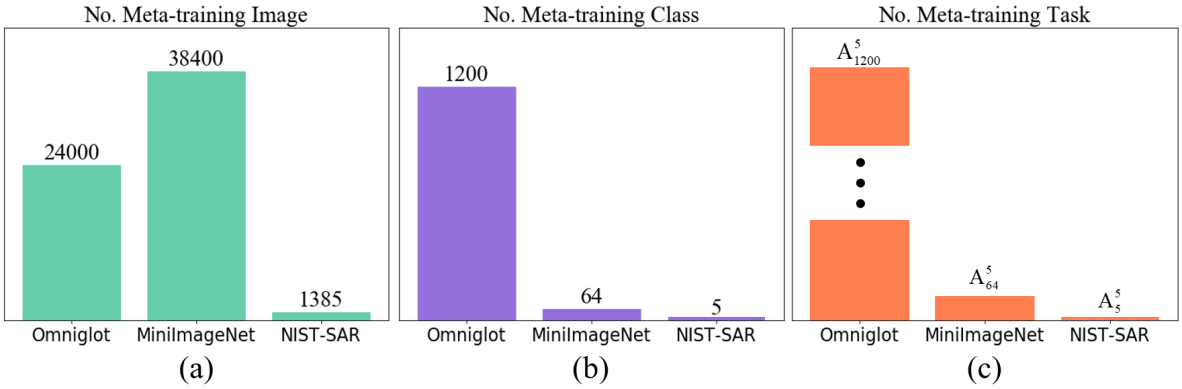


Fig. 9. Comparison of some meta informations on Omniglot, MiniImageNet and NIST-SAR data sets. (a)–(c) Different number of metatraining image, class, and task on three data sets, respectively.

TABLE IV

SETTING OF THE COMPARISON EXPERIMENTS WITH DIFFERENT METATRAINING CLASSES ON THE MINIIMAGENET DATA SET. WE CONTROL THE NUMBER OF THE METATRAINING IMAGES PER CLASS FOR FAIR COMPARISON

Few-task experiments	Exp_{64}	Exp_{32}	Exp_{16}	Exp_{10}	Exp_5
No. Meta-training Class	64	32	16	10	5
No. Meta-training Image (per class)	50	100	200	300	600
No. Meta-training Image (all)	3200	3200	3200	3200	3000
No. Meta-training Task	9.14×10^8	2.42×10^7	5.24×10^5	30240	120

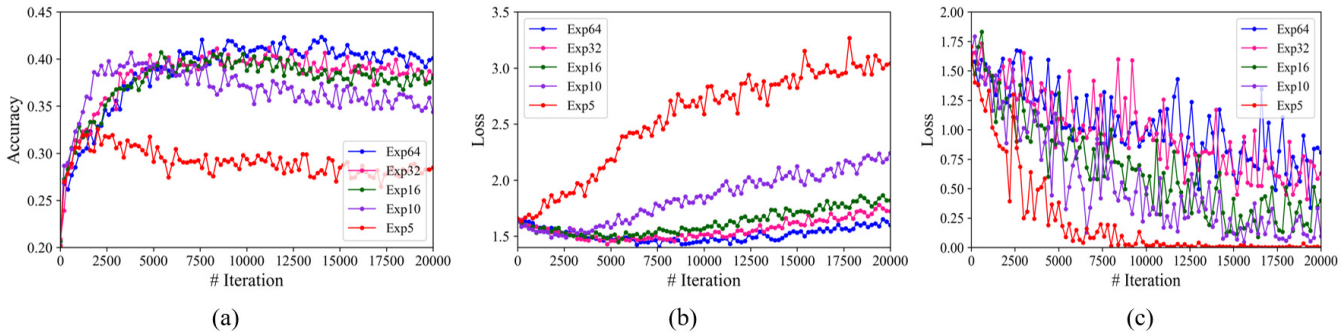


Fig. 10. (a) Metatest accuracy curve, (b) metatest loss curve, and (c) metatraining loss curve of the comparison experiments with different metatraining classes on the MiniImageNet data set. Best viewed in color.

Besides, a hard task mining method can further promote effective metalearning. We analyze the effect of the above two methods by the following ablation study.

1) Effectiveness of TO: The effectiveness of TO is evaluated on different few-shot SAR target classification tasks in different few-shot settings. As shown in Table III, the experiments demonstrate that MSAR_{TO} with the proposed TO can achieve significant improvements (0.4% accuracy gain for 1-shot and 9.7% accuracy gain for 5-shot) over the baseline method MSGD. Besides, we also visualize the metatest accuracies every 200 metatraining episodes in Fig. 7, we can see that when the results of our model (MSAR_{TO}) with TO become steady, the performances of our model are always better than baselines, and the training processes of MSAR_{TO} are more stable than baselines, which indicates that TO can avoid overfitting and leverage the pretrained prior knowledge effectively. Therefore, TO is a feasible method to address the few-task problem.

2) Different Settings for TO: We further analyze the performance of TO with two different settings. One is to set ϕ_w for each channel of all convolutional filters in the base-learner (denoted as $\text{TO}_{\text{channel}}$, ϕ_w for every convolutional layer is 64×64), another is to set one learnable transfer parameter for each convolutional filter (denoted as $\text{TO}_{\text{filter}}$, ϕ_w for every convolutional layer is 64×1). The comparison experiments are run in the 5-way, 1-shot and 5-way, 5-shot settings, and the results are shown in Fig. 11. On the NIST-SAR data set, $\text{TO}_{\text{filter}}$ can yield better performance with a more suitable number of learning parameters. But it should be stressed that the parameter setting for TO may vary from data set to data set.

3) Hard Task Mining: The proposed hard task mining method is simple and effective to promote effective

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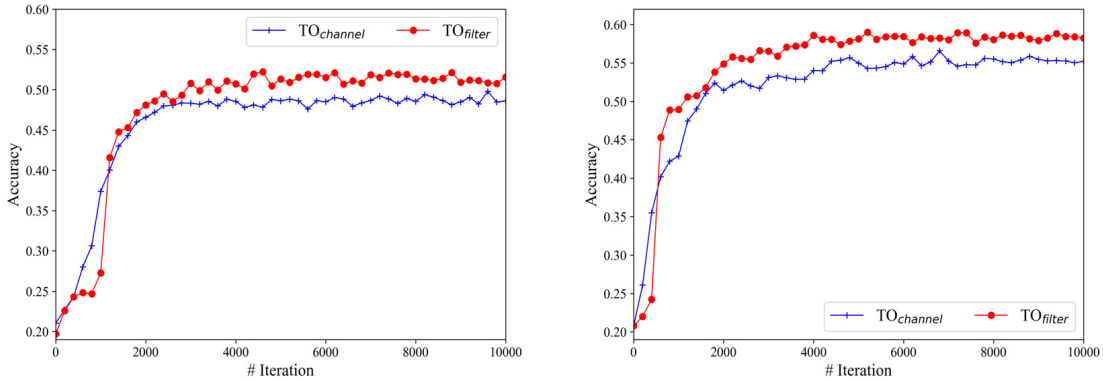


Fig. 11. Experiment results of different TO settings. (a) For 1-shot. (b) For 5-shot.

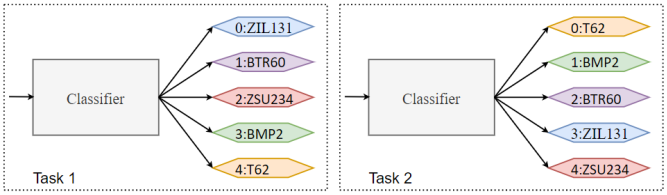


Fig. 12. Two different types of tasks. Best viewed in color.

metalearning. In this experiment, for a batch of tasks, we simply view the task with the highest loss value on $\mathcal{T}^{\text{te}}$ as the hard task and retrain it in the next batch. The hard task mining is only used for the metatraining phase. With this method, the proposed model yields higher accuracies of 53.5% for 1-shot and 60.5% for 5-shot, shown in Table III.

G. Mutual Exclusion Between Tasks

We take the mutual exclusion between tasks into account. Although there are only five classes for metatraining and metatest, respectively, we can build various tasks by randomly selecting target classes. In this article, we define the tasks with different arrangements of SAR target categories as different types. For example, as shown in Fig. 12, a task with classes (0:ZIL131, 1:BTR60, 2:ZSU234, 3:BMP2, 4:T62) is different from a task with classes (0:T62, 1:BMP2, 2:BTR60, 3:ZIL131, 4:ZSU234). Despite the two tasks have the same five categories, some of these categories have different labels. For ZIL131, its labels in two tasks are predicted with different positions on the predictions of the classifier. Therefore, the two tasks are mutually exclusive.

As mentioned in Section III-B, five target classes can construct $A_5^5 = 120$ types of tasks. The probability of no repeated task types in a batch (the batch size of tasks is 4 in our experiment) is $A_{120}^4/120^4 \approx 0.9508$. We select 48 tasks of 12 batches from our 5-way, 1-shot experiment and provide the detailed target categories for each of the 48 tasks in Fig. 13, most of the 48 tasks are mutually exclusive, and the tasks in each batch are of different types. Based on the above analysis, the tasks are usually mutually exclusive in a batch, and the tasks of the same type are not contiguous during metatraining or metatesting due to randomly selecting target classes. Therefore, mutual exclusion between tasks can be guaranteed.

Task \ Label	0	1	2	3	4
0	BMP2	ZIL131	T62	BTR60	ZSU234
1	BMP2	ZSU234	BTR60	ZIL131	T62
2	ZIL131	BTR60	BMP2	T62	ZSU234
3	ZIL131	ZSU234	BMP2	T62	BTR60
4	T62	ZIL131	ZSU234	BTR60	BMP2
5	ZIL131	T62	BTR60	ZSU234	BMP2
6	ZIL131	BTR60	ZSU234	BMP2	T62
7	ZIL131	ZSU234	T62	BMP2	BTR60
8	BMP2	T62	ZSU234	BTR60	ZIL131
9	BTR60	ZSU234	BMP2	T62	ZIL131
10	ZIL131	BTR60	T62	ZSU234	BMP2
11	T62	ZIL131	BTR60	ZSU234	BMP2
12	ZSU234	ZIL131	BMP2	T62	BTR60
13	ZIL131	BTR60	ZSU234	BMP2	T62
14	T62	BMP2	BTR60	ZIL131	ZSU234
15	BTR60	ZIL131	T62	ZSU234	BMP2
16	BMP2	T62	ZIL131	ZSU234	BTR60
17	T62	ZIL131	ZSU234	BTR60	BMP2
18	ZIL131	BTR60	T62	BMP2	ZSU234
19	BMP2	ZIL131	ZSU234	BTR60	T62
20	ZSU234	T62	BTR60	ZIL131	BMP2
21	BTR60	T62	ZSU234	ZIL131	BMP2
22	BTR60	ZSU234	T62	ZIL131	BMP2
23	T62	ZIL131	BMP2	ZSU234	BTR60
24	ZIL131	T62	BMP2	ZSU234	BTR60
25	ZIL131	T62	BTR60	BMP2	ZSU234
26	T62	ZIL131	ZSU234	BMP2	BTR60
27	BTR60	T62	ZIL131	BMP2	ZSU234
28	T62	BMP2	ZSU234	ZIL131	BTR60
29	T62	BTR60	ZIL131	BMP2	ZSU234
30	BMP2	BTR60	T62	ZIL131	ZSU234
31	ZIL131	T62	ZSU234	BTR60	BMP2
32	ZSU234	ZIL131	BMP2	BTR60	T62
33	BTR60	T62	ZSU234	BMP2	ZIL131
34	ZSU234	ZIL131	T62	BTR60	BMP2
35	T62	BMP2	ZSU234	BTR60	ZIL131
36	ZSU234	BMP2	ZIL131	BTR60	T62
37	ZIL131	T62	BTR60	BMP2	ZSU234
38	T62	ZSU234	ZIL131	BMP2	BTR60
39	ZSU234	ZIL131	BMP2	BTR60	T62
40	ZSU234	BMP2	ZIL131	BTR60	T62
41	T62	BTR60	ZSU234	BMP2	ZIL131
42	BTR60	ZIL131	ZSU234	BMP2	T62
43	ZIL131	BMP2	T62	BTR60	ZSU234
44	T62	BTR60	BMP2	ZIL131	ZSU234
45	BTR60	BMP2	ZSU234	ZIL131	T62
46	T62	BMP2	BTR60	ZSU234	ZIL131
47	T62	BTR60	ZIL131	ZSU234	BMP2

Fig. 13. Target categories for each of the 48 tasks. The batch size of tasks is 4. Different colors correspond to different categories. Best viewed in color.

V. Conclusion

This work proposes to solve the few-shot SAR ATR problem with metalearning and devises the NIST-SAR data set for effective training and evaluation of metalearning ATR methods. We define the few-task problem by analyzing the effect of training classes on the performance of metalearning models. This problem may be inevitable in some metalearning

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applications with few available target categories, especially for SAR ATR. Three transfer-learning methods are introduced to alleviate the difficulties of metalearning by leveraging the prior knowledge from the pretraining phase. Besides, the metalearning model can be trained more effectively with the proposed hard task mining method. This work proves the feasibility of metalearning for few-shot SAR ATR by the promising experimental results and can inspire the relevant researches. We will continue the research on a few-shot SAR ATR in the future.

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